

A Measurement-Grounded Digital Twin for Evaluating Cooperative MARL in Energy-Aware 5G UPF Orchestration

Shima Afshar Borji*, Roberto Bruschi*[†], Chiara Lombardo*[†], Raffaele Bolla*[†], Cristina Emilia Costa[†]

*DITEN, University of Genoa, Genoa, Italy [†]CNIT – S2N National Lab, Genoa, Italy

shima.afshar.borji@edu.unige.it

{roberto.bruschi, chiara.lombardo, raffaele.bolla}@unige.it, ccosta@cnit.it

Abstract—The 5G User Plane Function (UPF) can run as a DPDK kernel-bypass stack, which offers high throughput at a high idle-power floor, or as a user-space (USR) stack, which draws less power but has a narrower safe operating envelope. Selecting the right realization per site, across a fleet of heterogeneous Mobile Edge Computing (MEC) clusters, is a cooperative control problem whose multi-site coordination structure has not been studied. We build a measurement-grounded digital twin calibrated on a physical testbed through RAPL power measurements and per-mode surrogate models, and use it to evaluate cooperative multi-agent reinforcement learning (MARL) for joint UPF realization switching across $K = 10$ clusters. MAPPO, with a shared-parameter actor and a centralised critic, outperforms an independent per-site PPO (IPPO) ensemble, a centralised single-MLP PPO, and the swept-best classical hysteresis baseline by 10 to 61% under a physics-grounded reward that folds measured switching energy into specific energy consumption. The win over IPPO is small in magnitude but statistically robust (paired bootstrap one-sided $p = 0.009$, 95% CI [+28, +262] reward units), and it concentrates on the data-poor low-load clusters where independent learning lacks cross-site signal. The result indicates that coordination architecture matters more than algorithm choice for energy-aware UPF orchestration at the edge.

Index Terms—5G User Plane Function, energy efficiency, cooperative multi-agent reinforcement learning, MAPPO, digital twin, mobile edge computing, CTDE.

I. INTRODUCTION

The proliferation of 5G base stations and core network functions is driving a sharp increase in telecommunications energy consumption. The User Plane Function (UPF), which is responsible for packet routing, forwarding, and QoS enforcement in the 5G Core, is the highest-throughput data-plane element in a typical mobile operator deployment and one of its largest energy consumers at the network edge [1]. Operators face growing pressure to reduce their carbon footprint while maintaining stringent QoS commitments across a geographically distributed fleet of MEC clusters.

Two UPF realization classes define the practical energy-performance design space. A DPDK-accelerated kernel-bypass stack delivers high throughput and low tail latency, at the cost of dedicated CPU cores held at an elevated idle-power floor regardless of the offered load. A user-space (USR) software stack draws substantially less power at low utilization, but it has a narrower safe operating envelope before delay and

packet loss degrade below QoS targets [2]. Static all-DPDK provisioning, which is the current industry default, wastes energy whenever traffic falls below the per-site break-even point, while static all-USR violates QoS during peaks. The question is when to run which realization, on a per-site basis, given noisy short-horizon traffic forecasts and non-negligible switching costs.

Hysteresis-based switching provides the strongest classical answer. Upper and lower thresholds derived from the energy-performance crossover point gate transitions and prevent rapid oscillation near the decision boundary. At a single site this approach achieves meaningful energy savings [3], yet it faces a fundamental limit at practical forecaster accuracy levels. The hysteresis band must scale with the forecaster mean absolute error (MAE) to filter noise reliably, but when the MAE is large relative to the decision threshold, the lower threshold collapses to zero and the controller never leaves DPDK regardless of actual load. Single-site reinforcement learning (RL) resolves this by learning a policy that handles forecast uncertainty directly [4], but when each of K heterogeneous MEC clusters trains independently, data-poor low-load sites lack sufficient sample signal to discover that USR is safe at their operating point. Independent per-site learning leaves each cluster’s policy bounded by its own experience, with no mechanism to transfer the “USR-at-low-load” rule learned on data-rich clusters to the low-traffic sites that need it most.

The natural remedy is cooperative multi-agent training. A shared-parameter actor trained simultaneously across all clusters can propagate policies across sites, while decentralised execution preserves per-site autonomy at inference time. Whether this coordination benefit materialises in practice, and whether it is statistically significant relative to independent per-site learning, has not been quantified for UPF orchestration. Answering this question requires a simulation environment whose fidelity to real hardware is established by measurement, so that results are grounded in actual energy-performance trade-offs rather than synthetic assumptions.

This paper makes three contributions.

- *A measurement-grounded digital twin for multi-site UPF evaluation.* We calibrate a digital twin on a physical testbed (Keysight LoadCore traffic generation, RAPL-via-Scaphandre power measurement, SD-Core DPDK UPF

and OAI USR UPF) and extend the single-site framework of [4] to $K = 10$ heterogeneous MEC clusters with distinct traffic profiles and per-mode MLP surrogate models for power and QoS.

- *A physics-grounded reward that unifies switching and steady-state energy.* We remove the redundant flat switching-cost constant of prior formulations [4] and fold the measured activation spike energy directly into the specific energy consumption (SEC) term, so switching costs emerge from testbed-measured activation durations rather than hand-set constants.
- *Evidence that coordination architecture matters more than algorithm.* A controlled three-architecture comparison at eight training seeds, between MAPPO with a shared-parameter actor and centralised critic [5], an IPPO ensemble [6], and a centralised single-MLP PPO, shows that the MAPPO architectural win over IPPO is significant at $p = 0.009$ (paired bootstrap, 95% CI $[+28, +262]$ reward units), with the gain concentrating on low-load clusters where independent learning lacks sufficient signal.

The remainder of the paper is organised as follows. Section II reviews related work. Section III describes the system model and digital twin. Section IV formalises the multi-site control problem and reward. Section V details the MARL architectures and the evaluation protocol. Section VI presents and analyses results, and Section VII concludes.

II. RELATED WORK

Rule-based UPF realization selection. Among classical strategies for dynamic network function management, hysteresis-based switching is the strongest and most practically deployable. A hysteresis policy defines an upper threshold t_{up} and a lower threshold $t_{down} = t_{up} - b$ around the energy-performance crossover point. A switch to the low-power realization is triggered only when predicted traffic falls below t_{down} , and a switch back occurs only above t_{up} . The band b acts as a noise filter that prevents rapid oscillation around the decision boundary. Simpler threshold policies, which switch on a single crossing without hysteresis, are strictly inferior, since they are susceptible to every forecaster error and incur excessive switching cost. This makes hysteresis the correct classical baseline rather than the bare threshold. Our prior work [3] studies this approach at a single site using polynomial surrogate models and an LSTM forecaster, achieving 10.46% energy savings over static DPDK provisioning. However, hysteresis faces a fundamental trade-off at real forecaster accuracy levels. The band b must be set proportionally to the forecaster MAE to filter noise reliably, yet this same scaling causes t_{down} to collapse towards zero when the MAE is large relative to the decision threshold, at which point the controller never leaves DPDK regardless of the actual load. We derive both thresholds directly from our calibrated digital twin and sweep the band over five values to identify the strongest classical operating point, so that the comparison is fair and not based on hand-tuned parameters (Section VI). UPF placement and chaining

across the network have been studied from an optimisation perspective [7], but the per-site realization-selection problem with heterogeneous energy-performance profiles and practical forecaster noise has received less attention.

Reinforcement learning for single-site UPF orchestration.

RL has been applied to UPF realization selection at a single site. Bellin et al. [8] use deep RL to choose among UPF deployment options based on power consumption at the 5G edge. Our prior work [4] formulates the DPDK/USR selection problem as a single-site Markov decision process trained on a measurement-grounded digital twin, achieving 16.9% energy savings over static DPDK provisioning with a physics-grounded reward. These single-agent formulations leave the multi-site coordination problem open, because each site learns independently and data-poor low-load clusters receive no benefit from the richer experience of high-load peers. The present paper extends the twin-based framework to $K = 10$ heterogeneous clusters and directly quantifies how much co-ordination across sites is worth.

Multi-agent RL for network resource management.

Co-operative MARL has been applied to network slicing, resource allocation, and traffic steering in MEC environments [9]. Value-decomposition methods such as QMIX [10] and VDN [11] impose a monotonic factorisation of the joint value function, which simplifies credit assignment but introduces an inductive bias poorly suited to the heterogeneous reward scales of our $K = 10$ deployment. Moreover, both are off-policy and require replay buffers, which add tuning complexity that conflicts with the episodic, short-horizon structure of the 15-minute UPF control problem. On-policy actor-critic methods, specifically the MAPPO architecture [5] with a shared-parameter actor and a centralised critic, provide Centralised Training with Decentralised Execution (CTDE) without these constraints and have shown competitive performance in cooperative discrete-action tasks. We adopt MAPPO as our proposed architecture and include an IPPO ensemble [6] and a centralised single-MLP PPO as the two natural ablation points.

Digital twins for network control evaluation. Measurement-grounded digital twins have emerged as a principled approach to training and evaluating network controllers without incurring the cost and risk of live closed-loop experimentation [4], [14]. The fidelity of the twin is critical, since a surrogate that does not capture the energy-performance trade-offs of the underlying hardware will produce controllers that misfire at deployment. We address this by calibrating the twin on per-mode profiling data from a physical testbed and grounding activation durations and power curves in RAPL measurements [15], extending the single-site calibration of [4] to a ten-cluster heterogeneous fleet.

III. SYSTEM MODEL AND DIGITAL TWIN

A. Deployment and control loop

A mobile operator runs $K = 10$ MEC clusters that serve distinct geographic regions, each hosting one UPF instance. Fig. 2 shows where the controller sits. It operates at the

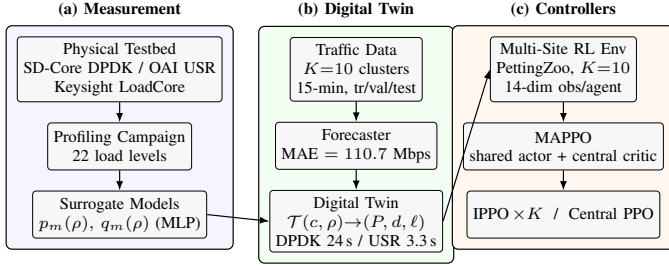


Fig. 1. End-to-end pipeline. (a) Physical-testbed measurements feed the profiling campaign that produces per-mode MLP surrogate models; (b) per-cluster traffic and a short-horizon forecaster are combined with the surrogates inside the measurement-grounded digital twin; (c) the twin drives a multi-site PettingZoo RL environment shared by the three evaluated controllers.

management and orchestration plane, above the 3GPP control plane, and issues virtual network function (VNF) lifecycle operations that start or stop the DDPK and USR containers at each site. The 5G Core control plane (SMF, AMF, UDM, PCF) stays centralised, and the SMF keeps issuing N4/PCF session rules to whichever realization is active at each site. At each control step t on a 15-minute cadence, cluster i chooses a realization $a_{i,t} \in \{0, 1\}$, where 0 denotes DDPK and 1 denotes USR. Switching is allowed at every step, subject to a cost. The per-agent action space is binary and the joint action space is $\{0, 1\}^K$, which grows as 2^K . The control loop is implemented as a PettingZoo parallel environment [16] wrapped around the twin.

B. Digital twin

We use a calibrated digital twin $\mathcal{T}$ that returns, for each cluster i and step t , given an offered load $\lambda_{i,t}$ and a realization $a_{i,t}$, the steady-state power $P_{i,t}^{\text{steady}}$ in watts, the per-packet delay $d_{i,t}$ in microseconds, the predicted packet loss $\ell_{i,t}$, and a binary safety flag. The twin is built in three steps that match Fig. 1. First, a physical testbed (SD-Core DDPK UPF, OAI USR UPF, Keysight LoadCore traffic generation) is profiled over 22 load levels while RAPL-via-Scaphandre records package power. Second, per-mode multilayer-perceptron (MLP) surrogates $p_m(\rho)$ and $q_m(\rho)$ regress power and QoS as functions of normalised load ρ for each realization m . Third, the RAPL traces also fix the measured activation durations of the two realizations, which are $t_{\text{DDPK}}^{\text{act}} = 24$ s and $t_{\text{USR}}^{\text{act}} = 3.3$ s. These durations make the switching cost asymmetric and physically grounded rather than hand-set. We treat the surrogates as ground truth in this paper and defer sim-to-real validation to future work.

C. Traffic and forecaster

Per-cluster offered loads come from a short-horizon forecaster [4] and are partitioned chronologically into a training slice of 5073 steps (about 53 days), a validation slice of 1009 steps (about 10.5 days), and a test slice of 1009 steps (about 10.5 days). The one-step forecast $\hat{\lambda}_{i,t+1}$ is exposed to the controllers as part of the observation. The forecaster MAE on the test slice, averaged across the ten clusters, is 110.7 Mbps,

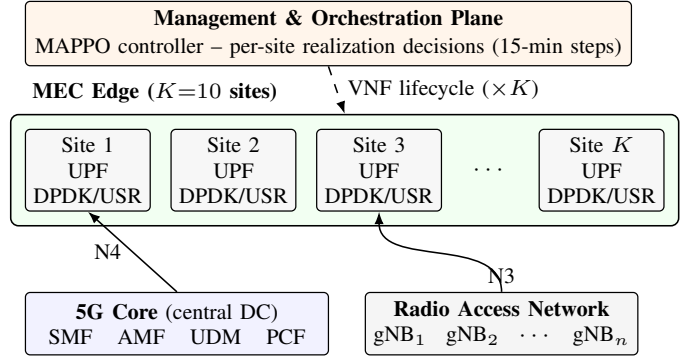


Fig. 2. 5G deployment model. The MAPPO controller operates at the management and orchestration plane and issues container start/stop operations to switch each MEC site between DDPK and USR realizations. The 5G Core control plane stays centralised, and the SMF continues to issue N4/PCF session rules to the active UPF at each site.

which is non-negligible relative to the per-site break-even point and is the reason naive hysteresis struggles (Section IV).

The ten clusters form a deliberately heterogeneous fleet, which is what makes coordination interesting. On the test slice the per-cluster mean load ranges from 0.108 Gbps (cluster c0) to 1.66 Gbps (cluster c9), a $15.4\times$ spread, with 95th-percentile loads from 0.22 to 3.43 Gbps and instantaneous peaks up to 5.8 Gbps. The clusters also differ in shape, with peak-to-trough ratios between 4.8 and 17.3 and a daily peak that falls between 10:00 and 11:00 for most sites. This spread means a single static policy cannot be efficient everywhere, since the lowest-load clusters spend most of the day below the USR break-even point while the highest-load clusters are DDPK-bound around the clock.

IV. MULTI-SITE CONTROL PROBLEM

A. Per-site observation and objective

Each agent observes a 14-dimensional state at step t , formed by the current load $\lambda_{i,t}$, an 8-step history window $\lambda_{i,t-1}, \dots, \lambda_{i,t-8}$, the one-step forecast $\hat{\lambda}_{i,t+1}$, the previous action $a_{i,t-1}$, the previous QoS score $Q_{i,t-1}$, the previous specific energy $\text{SEC}_{i,t-1}$, and a normalised cooldown-progress signal. The cooperative objective is the discounted sum of per-cluster rewards over an episode of length $T = 1009$,

$$J(\pi) = \mathbb{E}_{\pi} \left[\sum_{t=0}^{T-1} \gamma^t \sum_{i=1}^K r_{i,t} \right], \quad \gamma = 0.995. \quad (1)$$

Coupling across agents enters only through the joint cumulative reward. Per-cluster dynamics are independent given the realization choice, and we defer shared-resource extensions such as a common CPU pool to future work.

B. Physics-grounded reward

Following the single-site formulation of [4] but removing its parallel flat switching constant, the per-step reward is

$$r_{i,t} = -(\alpha \text{SEC}_{i,t} + \lambda_{\text{QoS}} \max(0, \tau - Q_{i,t}) + L_{i,t}^{\text{CD}}). \quad (2)$$

The specific energy term now carries both steady-state and transition power. With the twin’s per-transition spike energy $E_{i,t}^{\text{sw}} = P_{\text{new}}(\lambda_{i,t}) t_{\text{new}}^{\text{act}}$ and a step duration $\Delta t = 900$ s, we define

$$P_{i,t}^{\text{eff}} = P_{i,t}^{\text{steady}} + E_{i,t}^{\text{sw}} / \Delta t, \quad (3)$$

$$\text{SEC}_{i,t} = P_{i,t}^{\text{eff}} / \max(\lambda_{i,t}, \varepsilon) \quad [\text{W/Mbps}]. \quad (4)$$

The QoS score $Q_{i,t} = \min(\text{score}_d(d_{i,t}), \text{score}_\ell(\ell_{i,t})) \in [0, 1]$ grades delay and loss against their budgets, and $L_{i,t}^{\text{CD}}$ is a soft anti-oscillation penalty that fires when a switch happens inside a cooldown window. The constants are $(\alpha, \lambda_{\text{QoS}}, \tau) = (100, 30, 0.9)$, with cooldown period 4 steps and cost $c_{\text{CD}} = 0.5$, matching the paper-aligned values of [4]. The asymmetry between DPDK and USR switching costs now emerges automatically from $t_{\text{DPDK}}^{\text{act}} \gg t_{\text{USR}}^{\text{act}}$ and from load scaling, since a DPDK activation under heavy traffic costs more than one under light traffic, as it physically should. A previous revision used flat per-event constants running in parallel to the twin’s E^{sw} , and we removed that redundancy.

C. Classical baseline derivation

The threshold and hysteresis baselines use operating points derived from the twin rather than hand-tuned. Sweeping the twin gives an energy break-even at 91.0 Mbps, where USR power first exceeds DPDK power, and a QoS limit at 149.0 Mbps, the last load where USR stays safe. Taking the minimum and subtracting a 10 Mbps safety margin yields a decision threshold of 81.0 Mbps. A paper-faithful hysteresis band set to twice the forecaster MAE is 221.3 Mbps, which is more than $2.7\times$ the decision threshold. As a result the lower threshold collapses to zero and the auto-tuned hysteresis controller never leaves DPDK after the first step. This is the honest behaviour of naive hysteresis at this forecaster accuracy, not a bug. To give the classical family its strongest possible form, we additionally sweep a narrower band over $\{5, 10, 20, 50, 100\}$ Mbps and report the best operating point (Section VI).

V. COOPERATIVE MARL CONTROLLERS

MAPPO (proposed). MAPPO [5] uses a parameter-shared actor, an MLP $14 \rightarrow 64 \rightarrow 64 \rightarrow 2$ with 5,250 parameters in which one network handles every agent’s per-step decision, together with a centralised critic on the global state, an MLP $140 \rightarrow 128 \rightarrow 128 \rightarrow K$ with 35,850 parameters. The global state is the concatenation of all per-agent observations ($K \times 14 = 140$). Training uses PPO clipping [12] on per-agent ratios with GAE [13] ($\lambda = 0.9$), per-agent advantage normalisation over (T, K) to absorb the cross-cluster reward-scale differences, and reward scaling of 0.01 to balance actor and critic losses. We run 200k environment steps with $n_{\text{steps}} = 1024$, $n_{\text{epochs}} = 10$, minibatch 256, learning rate 3×10^{-4} , and entropy coefficient 0.05. Best-of-validation checkpoint selection is load-bearing, because without it a late-stage policy collapse on a near-zero-load step produces an unusable final-step model.

TABLE I
TEST-SLICE COMPARISON ($K = 10$, 1009-STEP EPISODE). MAPPO AND IPPO AT 8 SEEDS, CENTRALISED PPO AT 4. REWARD IS THE UNWEIGHTED SUM OVER CLUSTERS AND STEPS; QoSV IS THE QoS-VIOLATION RATE.

Controller	Reward n	Wh	QoSv%	USR%	
MAPPO (ours)	-5633 ± 100	8	1967	0.52	7.8
IPPO ensemble	-5787 ± 89	8	1973	0.57	8.1
Hysteresis ($b=50$)	$-6262 -$	1995	0.68	7.5	
Hysteresis ($b=20$)	$-6543 -$	1971	0.94	11.5	
Always-DPDK	$-7201 -$	2071	0.38	0.0	
Threshold (81 Mbps)	$-7623 -$	1970	1.42	14.0	
Centralised PPO	$-14,458 \pm 4509$	4	2130	3.73	9.3

IPPO ensemble (ablation). The IPPO ensemble trains $K = 10$ independent Stable-Baselines3 [6] PPO actors, one per cluster, on each cluster’s 14-dimensional observation with identical hyperparameters. At evaluation the per-cluster policies are stacked. By construction the ensemble consumes K times the training budget, which is its nominal compute advantage over the shared-actor design.

Centralised PPO (ablation). The centralised baseline is a single SB3 PPO on the joint 140-dimensional observation with a MultiDiscrete($[2]^K$) joint action, otherwise matched in reward, budget, and checkpoint selection.

Evaluation protocol. Each trained controller is evaluated once on the test slice with a deterministic policy and a fixed environment seed for reset, over the full 1009-step episode on every cluster. MAPPO and IPPO are run at eight seeds $\{1, 7, 13, 23, 42, 64, 77, 99\}$, and centralised PPO at four seeds, since its failure mode is unambiguous at any sample size. Classical baselines are deterministic. We report reward as the unweighted sum over clusters and steps, total fleet energy in watt-hours, the QoS-violation rate (the fraction of cluster-steps whose delay or loss budget is exceeded), the USR usage share, and the switch count.

VI. RESULTS

A. Headline comparison

Table I reports test-slice means across all controllers. MAPPO attains the lowest reward magnitude, the lowest energy, and the lowest QoS-violation rate among the trained policies. On reward it improves on the swept-best classical baseline (hysteresis at band 50 Mbps) by 10.0%, on always-DPDK by 21.8%, and on the centralised PPO by 61.0%, which is the 10 to 61% range quoted in the abstract. The margin over the IPPO ensemble is much smaller in magnitude (154 reward units, about 2.7%) but is statistically robust, as the next subsection shows. MAPPO is also the lowest-energy trained controller at 1967 Wh, ahead of IPPO at 1973 Wh and well ahead of always-DPDK at 2071 Wh. The energy gap to IPPO is small, so the reward advantage comes less from raw watt-hours and more from better-placed USR usage that avoids QoS penalties (Section VI-C).

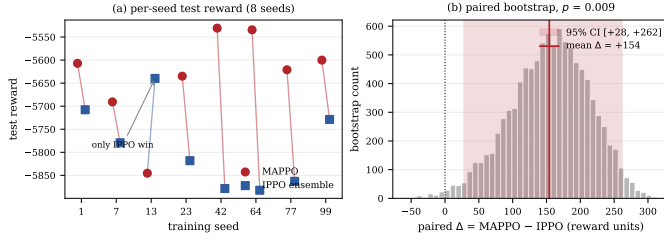


Fig. 3. (a) Per-seed test reward for MAPPO (circles) and the IPPO ensemble (squares) over eight seeds, with paired lines connecting seed-matched runs. Seed 13 is the only seed where IPPO wins. (b) Paired-bootstrap distribution (10,000 resamples) of the mean $\Delta = \text{MAPPO} - \text{IPPO}$ delta, with the 95% CI [+28, +262] excluding zero ($p = 0.009$).

B. Statistical significance

A paired bootstrap over the eight seed-matched MAPPO minus IPPO deltas (10,000 resamples) places the architectural gap at +154 reward units, with a 95% CI of [+28, +262] and a one-sided $p = 0.009$. The interval excludes zero by a clear margin, so the MAPPO win over IPPO is significant at the 0.01 level under the physics-grounded reward. Fig. 3(a) shows that seven of the eight seeds favour MAPPO. The single counter-example is seed 13, where MAPPO converged to a slightly more aggressive USR policy (0.67% QoS violation, 364 switches, against the MAPPO averages of 0.52% and 297) and IPPO happened to train its single best ensemble of the set. Fig. 3(b) shows the bootstrap distribution. Both architectures have comparable seed-to-seed reward range (MAPPO 315, IPPO 243), so MAPPO is slightly less stable across seeds but reliably better on the mean.

C. Cross-cluster transfer

Fig. 4 breaks the test reward down by cluster, with clusters sorted by mean offered load. MAPPO wins on 7 of 10 clusters, ties on the high-load DDPK-bound cluster c3, and loses by a small margin on two mid-load clusters (c2 by 8.8% and c8 by 0.4%). The clearest wins are on the low- and mid-load clusters, c5 (+8.6%), c0 (+6.6%), c7 (+4.3%), and c4 (+3.4%), which are exactly the sites where USR is the right call but a per-cluster IPPO has limited sample signal to find it. On c0, the lowest-load cluster, MAPPO’s shared actor drives the most USR usage and the largest gap to always-DPDK. The shared actor, trained simultaneously on the data-rich low-load clusters, transfers the “USR-at-low-predicted-load” rule to sites where an independent learner would stay close to DDPK. The small regression on c2 is the cost of policy sharing on a heterogeneous fleet, where MAPPO occasionally tries USR on a cluster that an independent learner has correctly fixed to DDPK, and it is more than offset by the low-load wins. On the highest-load cluster c9, with peaks near 5.8 Gbps, a few IPPO seeds tried USR and were punished, leaving the IPPO mean slightly below always-DPDK, whereas MAPPO correctly stays on DDPK across all eight seeds.

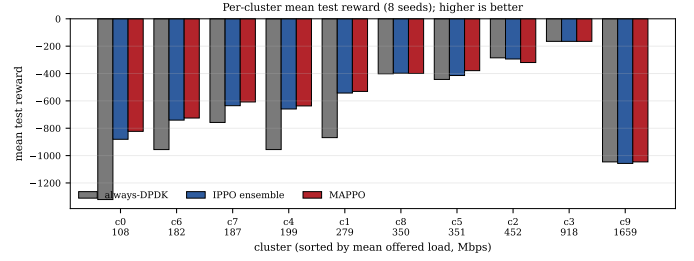


Fig. 4. Per-cluster mean test reward (8 seeds) for always-DPDK, the IPPO ensemble, and MAPPO, with clusters sorted by mean offered load (Mbps). MAPPO’s advantage concentrates on the low- and mid-load clusters, where independent learning lacks the signal to leave DDPK. The high-load clusters c3 and c9 are DDPK-bound for every controller.

TABLE II
HYSTERESIS BAND SWEEP ($t_{up} = 81$ MBPS, COOLDOWN 1 STEP). BAND 50 MBPS IS THE STRONGEST CLASSICAL OPERATING POINT, STILL 629 REWARD UNITS BEHIND MAPPO.

Band	t_{down}	Reward	Switches	QoS%	Regime
5	76	-7047	317	1.16	over-switching
10	71	-6923	285	1.11	over-switching
20	61	-6543	215	0.94	prior hand-pick
50	31	-6262	111	0.68	best classical
100	0	-7201	0	0.38	collapses to DDPK

D. Classical-baseline sensitivity

Table II sweeps the hysteresis band over five widths, with t_{up} fixed at the derived 81 Mbps decision point and a one-step cooldown. Two facts stand out. First, the hand-picked 20 Mbps band used in earlier iterations is not the best classical operating point. Widening to 50 Mbps cuts switching from 215 to 111 events and improves reward by 281 units, while a band of 100 Mbps drives t_{down} to zero and collapses the controller to always-DPDK. Second, no setting of the band closes the gap to MAPPO. Even at the sweep optimum, hysteresis is 629 reward units behind MAPPO, far outside the MAPPO minus IPPO 95% CI and any per-seed range. The classical controller’s fundamental limit at this forecaster accuracy is that any band wide enough to suppress noise-induced switching either degenerates to always-DPDK or admits a non-trivial QoS-violation rate.

A complementary cooldown sweep on MAPPO (seed 42) around the default period 4 and cost 0.5 confirms the headline does not hinge on these hyperparameters. Lowering the cost to 0.1 lets the controller switch 58% more often and costs 526 reward units, while raising it to 1.0 overpenalises switching and costs 264 units. Varying the period between 2 and 6 steps moves reward by at most 144 units. The default sits in the centre of a soft plateau whose width is on the order of the MAPPO minus IPPO gap and well inside the gap to the strongest classical baseline.

E. Why centralised PPO fails

The centralised single-MLP PPO sees the same problem, reward, and training budget, yet it trails MAPPO by 61% with

a seed-to-seed range of about 9,930 reward units (per-seed rewards $-11,202$, $-12,718$, $-12,779$, $-21,132$). Its energy at 2130 Wh is worse than always-DPDK, because it spends transition energy without the offsetting low-load USR benefit. Inspecting per-cluster behaviour reveals the failure mode. The joint factored softmax collapses one or two clusters to mostly-USR even when their load is unsafe for it, for example one seed drives cluster c6 to 98% USR with a 46% QoS-violation rate, and another drives c7 to 56% USR with a 23% violation rate. A single unsafe cluster of this kind dominates the joint reward and never recovers, because the 140-dimensional to 10-dimensional factored softmax cannot factor the per-cluster substructure within the training budget. The diagnosis is architectural rather than optimisational. Ten parameter-sharing 14-to-2 policies are a much easier optimisation than one joint policy over a 2^K action space.

VII. CONCLUSION

We cast per-site DPDK/USR realization switching across $K = 10$ heterogeneous MEC clusters as a cooperative MARL problem on a measurement-grounded digital twin, under a physics-grounded reward that folds the twin’s measured switching energy into specific energy consumption. On a 10.5-day held-out test slice, MAPPO with a shared-parameter actor and a centralised critic beats an IPPO ensemble, a centralised single-MLP PPO, and the swept-best classical hysteresis controller by 10 to 61% on reward. The win over IPPO is small in magnitude but statistically significant (paired bootstrap $p = 0.009$, 95% CI $[+28, +262]$), and it concentrates on the data-poor low-load clusters, which is consistent with cross-cluster policy transfer through the shared actor. The practical message is that for energy-aware UPF coordination at the edge, the choice of coordination architecture matters more than the choice of algorithm. Future work will hold out clusters at training time to test zero-shot transfer directly, add shared-resource coupling such as a common CPU pool, and validate the twin against a live DPDK/USR testbed.

REFERENCES

- [1] Y. Ding, H. Duan, M. Xie, J. Rao, J. J. Wang, and W. Zhang, “Carbon emissions and mitigation potentials of 5G base stations in China,” *Resources, Conservation and Recycling*, vol. 182, p. 106330, 2022.
- [2] J. Rischke, C. Vielhaus, P. Sossalla, J. Wang, and F. H. P. Fitzek, “Comparison of UPF acceleration technologies and their tail-latency for URLLC,” in *Proc. IEEE NFV-SDN*, 2022, pp. 146–151.
- [3] S. Afshar Borji, R. Bruschi, C. E. Costa, and C. Lombardo, “Dynamic energy-efficient User Plane Function selection in 5G networks,” in *Proc. IFIP SLICES Workshop*, 2025.
- [4] S. Afshar Borji, R. Bruschi, C. Lombardo, R. Bolla, and C. E. Costa, “AI-driven energy-efficient orchestration of User Plane Functions in 5G Core networks,” *Computer Communications*, 2026, under review, manuscript COMCOM-S-26-00430.
- [5] C. Yu, A. Velu, E. Vinitzky, J. Gao, Y. Wang, A. Bayen, and Y. Wu, “The surprising effectiveness of PPO in cooperative multi-agent games,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 35, 2022.
- [6] A. Raffin, A. Hill, A. Gleave, A. Kanervisto, M. Ernestus, and N. Dornmann, “Stable-Baselines3: Reliable reinforcement learning implementations,” *J. Mach. Learn. Res.*, vol. 22, no. 268, pp. 1–8, 2021.
- [7] I. Leyva-Pupo and C. Cervelló-Pastor, “Efficient solutions to the placement and chaining problem of User Plane Functions in 5G,” *J. Netw. Comput. Appl.*, vol. 197, p. 103253, 2022.

- [8] A. Bellin, N. Di Cicco, D. Munaretto, and F. Granelli, “Power consumption-aware 5G edge UPF selection using deep reinforcement learning,” in *Proc. IEEE NFV-SDN*, 2024, pp. 1–6.
- [9] Y. Zhu et al., “Multi-agent deep reinforcement learning for UAV-assisted 5G network slicing: A comparative study of MAPPO, MADDPG, and MADQN,” *arXiv preprint arXiv:2512.03835*, 2024.
- [10] T. Rashid, M. Samvelyan, C. Schroeder de Witt, G. Farquhar, J. Foerster, and S. Whiteson, “QMIX: Monotonic value function factorisation for deep multi-agent reinforcement learning,” in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2018.
- [11] P. Sunehag et al., “Value-decomposition networks for cooperative multi-agent learning based on team reward,” in *Proc. Int. Conf. Auton. Agents Multi-Agent Syst. (AAMAS)*, 2018.
- [12] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal policy optimization algorithms,” *arXiv preprint arXiv:1707.06347*, 2017.
- [13] J. Schulman, P. Moritz, S. Levine, M. Jordan, and P. Abbeel, “High-dimensional continuous control using generalized advantage estimation,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2016.
- [14] P. Almasan, M. Ferriol-Galmés, J. Paillisse et al., “Network digital twin: Context, enabling technologies, and opportunities,” *IEEE Communications Magazine*, vol. 60, no. 11, pp. 22–27, 2022.
- [15] K. N. Khan, M. Hirki, T. Niemi, J. K. Nurminen, and Z. Ou, “RAPL in action: Experiences in using RAPL for power measurements,” *ACM Trans. Model. Perform. Eval. Comput. Syst.*, vol. 3, no. 2, 2018.
- [16] J. K. Terry et al., “PettingZoo: Gym for multi-agent reinforcement learning,” *arXiv preprint arXiv:2009.14471*, 2021.